

An E-Voting System with RSA Homomorphic Encryption

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study the RSA and ELGAMAL cryptosystem and the way it can be utilized to e-voting system.

Abstract— E-Voting is much faster than existing manual voting with independent of location limitation. But one of the deal breaker of this system is privacy and security. In this work we propose a secure electronic voting protocol based on homomorphic encryption which will fit for a huge scale of voting over online. Here, the voter will cast his ballot anonymously and can know his vote is really count or not. Our proposal includes the multiplicative encryption and cloud server where intermediate result will be stored. The protocol contains five distinct phases - registration, authentication, key generation, voting and counting with private cloud voting server.

Keywords— cloud computing, homomorphic encryption, RSA cryptography, accuracy.

I. INTRODUCTION

Elections are very popular in democratic country that involves third party to calculate the ballots. But now-a-days everyone wants to get rid of from trusted third party. Thus the term Electronic vote comes and it offers that authenticate person can vote from anywhere which is good news for disable and old people. But the major drawback of e-voting is vulnerability. Thus many cryptography researchers are looking at this scheme. E-voting must need to promise faster tabulation of ballot with accuracy, less chance of error and also less expensive.

Existing homomorphic e-voting scheme mainly based on a server which keeps an eye to the whole procedure. This act's as a central heart of the system. If it goes down or compromised than the whole system will be dead. Thus in this work we rely on cloud server based on multiplicative homomorphic to ensure anonymity, privacy and reliability.

II. RELEATED WORK

Yingming Zhao et al. [1] introduce an e-voting scheme depends on homomorphic encryption to achieve anonymity, privacy and reliability. But there are some issues that need to fix to achieve those goals like how to prevent the conspiracy. In [2] the authors proposed a mix type remote electronic voting based on zero- knowledge proof. Beside these we also

III. STAGES

The e-voting scheme has five distinct phases: registration, authentication, voting and counting along with public and private cloud voting server.

- a) *Registration*: By using the central biometric database of the government, this scheme verifies the eligibility of a voter. If everything works then the voter will give a username and password to the cloud.
- b) *Authentication*: Voter needs to be authenticated before casting their vote. Authentication can be done by username and password. In our system each voter can only vote once.
- c) *Key Generation*: This phase will generate distinct primes for each candidate. It also generates public key K_{pub} and private key K_{prv} pairs for voter as well. Only K_{pub} will be sent to the voter and everything else will be stored by the server.
- d) *Voting*: This stage voter will encrypt his message and public cloud will store and display the multiplication of cipher text and private cloud will perform decryption.
- e) *Counting*: Finally the private cloud will perform the calculation and send the result to the public cloud where it will be displayed.

IV. SECURITY PROPERTIES

An e-voting system must need to satisfy the following security properties:

1. *Confidentiality*: In e-voting system the privacy of a voter is paramount. It cannot be disclosed at any cost not only before the vote but also after the vote as well. It gives them the opportunity to freely decide their opinion.
2. *Authentication and Authorization*: The authentication part will be done by the public cloud and the voter.
3. *Integrity*: No one can be able to modify the result at any time. Integrity also ensures that only voters casted the vote and also a single vote tallied from a single voter.
4. *Verifiability*: A voter can able to verify whether his vote cast or not.

V. HOMOMORPHIC ENCRYPTION

The basic idea of homomorphic encryption is to do calculation without decrypting the cipher text. It allows computations on cipher text, thus produce an encrypted result. If this result decrypt, then it will match with the result if the same operations performed on the plain text. It is very popular in the cloud computing for securing the data. Basically two types of homomorphic schemes are available: Partial Homomorphic and Fully Homomorphic schemes. In partial homomorphic scheme we can do computation like addition and multiplication. On the other hand full homomorphic support most of the operations but it is relatively slow to compare with the first one.

A. MULTIPLICATIVE HOMOMORPHIC

Let, there are two message m_1 and m_2 and their corresponding cipher text are c_1 and c_2 . Thus $c_1 = e_k(m_1)$ and $c_2 = e_k(m_2)$. Therefore $d_k(c_1 \cdot c_2) = m_1 \cdot m_2$. Thus it can be treated as a black box, which takes encrypted messages, perform operation on these messages and return the encrypted text. Therefore we can say an algorithm $e()$ is said to be homomorphic if given $e_k(m_1)$ and $e_k(m_2)$ can obtain $m_1 \cdot m_2$ without decrypting $m_1 \cdot m_2$.

C. RSA (Multiplicative Homomorphism)

In 1978, Ron Rivest, Adi Shamir and Leonard Adleman invent a public key cryptosystem called RSA and it's widely recognized for secure computation. In RSA, two keys are used one for encryption called public key and another for decryption called private key.

Given $c_i = e(m_i) = m_i^e \mod n$

$$\begin{array}{rcl} c_1 & = & m_1^e \mod n \\ c_2 & = & m_2^e \mod n \\ \hline c_1 \cdot c_2 & = & m_1^e \cdot m_2^e \mod n = (m_1 \cdot m_2)^e \mod n \end{array}$$

Multiplicative RSA Encryption

RSA Key Generation

1. Choose two large primes p and q
2. Compute $n = p \cdot q$
3. Find Euler's totient function $\phi(n) = (p-1)(q-1)$.
4. Choose the public exponent e such that $e \in \{2, \dots, \phi(n)-1\}$ and $\gcd(e, \phi(n))=1$
5. Compute the private key such that $d \cdot e \equiv 1 \mod n$

Finally, return the $k_{pub}=(e,n)$

RSA Encryption

1. To compute cipher text c from a plaintext message m , find $c = m^e \mod n$

Return c

RSA Decryption

1. Because of the relationship between d and e , the algorithm correctly recovers the original message m , since

$$P = c^d \mod n = (m^e)^d = m^{ed} = m^1 = m \mod n$$

VI. E-VOTING

Figure 1 shows the top level view of our e-voting architecture. First and foremost voter needs to be register to the system before the voting. After the registration process voter needs to be authenticate by the server. Election commission will generate distinct primes (K_{pub} and k_{pr}) for each of the candidates and K_{pub} will send to the voter via public cloud. Then each voter will vote his favorite person through his public key and send the encrypted result to the public cloud. In this section public cloud perform the homomorphic encryption and send the encrypted result to the private cloud (or private server). At the next step this private cloud will perform the decryption and send the result to public cloud to display.

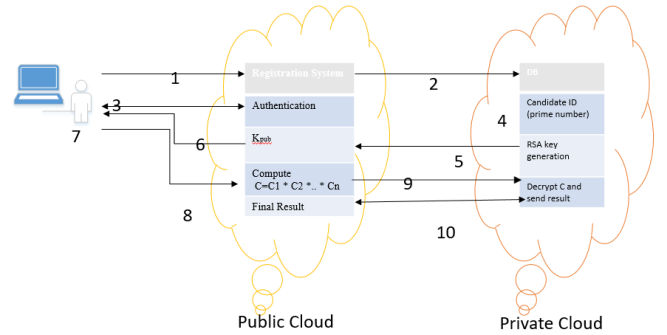


Figure 1: Level 0 architecture of e-voting system

A. Practical Use

Let assume, there are three voters who will going to cast their votes for two candidates. The private cloud will generate two distinct prime numbers for those two candidates (say 3 and 5). Now, RSA key generation phase execute and produce 11 and 17 for p , q respectively. Thus n will produce 187 and $\phi(n)$ returns 160. Let, public exponent is 7 thus private key is 23.

Now, let assume first and second voter voted for the first candidate. Thus $c_1 \equiv 3^7 \equiv 130$ and $c_2 \equiv 130$. Last voter voted for another candidate thus $c_3 \equiv 5^7 \equiv 146$. The public cloud will going to compute $C \equiv c_1 * c_2 * c_3 \equiv 130 * 130 * 146 \equiv 122$.

After encryption part decryption part comes. Now, by using private key ($d=23$) we compute $C^d \equiv (122)^{23} \equiv 45$. Now, we need to handle this result such as $45 = 3^2 * 5^1$. From where we can see the distinct prime for candidate one has exponent 2 and candidate two has 1 and those number indicate how many vote each candidate gets.

On the other hand if we assume the opposite scenarios where two voters votes for candidate two and one for candidate one than public cloud compute $C \equiv c_1 * c_2 * c_3 \equiv 146 * 146 * 130 \equiv 114$. After decryption we get $C^d \equiv (114)^{23} \equiv 75$. Thus using the algorithm we get candidate two has 2 in his exponent and candidate one has only one ($75 = 3^1 * 5^2$).

VII. VOTING ALGORITHM

Input: Candidate ID C_i , Public parameter n and e

Output: Result

Step1 Send a request to the public cloud server S_{pub} for registration and authentication.

Step2 After joining the network each voter will get the distinct prime number C_i for each candidate

Step3 Encrypt vote () in voter machine and send it to S_{pub}

Step4 S_{pub} process the vote and send it to private server S_{prv} .

Step4 S_{prv} Decrypt vote(), calculate() and display result ()

/* Calculate () : Initially $VC_i=0$

if ($PT \% C_1 == 0$)

{

$VC_1 = VC_1 + 1$;

$PT = PT / C_1$;

}

Else

{

if ($PT \% C_2 == 0$)

{

$VC_2 = VC_2 + 1$;

$PT = PT / C_2$;

}

We can summarize our work as following:

1. Due to there is only n (n number of voters) number of encryption and one single decryption our algorithm is comparatively faster than others.
2. Public cloud system will not hamper the security issue of the algorithms

VIII. WEAKNESS

If a voter does not verify the system after casting his vote, then the public cloud can cheat with the system. To understand this weakness let assume at initial stage voter V_1 vote for a candidate C_1 and encrypt his vote and send $E(C_1)$ to the public cloud. Here public cloud easily insert any cipher text, say $E(C_2)$ and can cheat. Another weakness is one voter can vote a lot of times for a specific candidate which can be caught at the last stage when private cloud calculate the total number of votes but cannot be solved by the system.

IX. E-VOTING IN SEVERAL COUNTRIES^[3]

Australia: In October 2001 electronic casting a ballot was utilized without precedent for an Australian parliamentary decision. In that race, 16,559 voters (8.3% of all votes tallied) cast their votes electronically at surveying stations in four spots. The Victorian State Government presented electronic casting a ballot on a preliminary reason for the 2006 State race.

India: Around 370 million Indian voters, (India has 675 million voters), submitted electronic votes in the all-electronic 2004 general race, One million electronic throwing tally

machines were scattered in a whole of 800 000 studying stations.

France: In January 2007 France's UMP party held a national presidential essential utilizing both remote electronic casting a ballot and with 750 surveying stations utilizing contact screen electronic casting a ballot over the Internet.

X. CONCLUSION

By using RSA multiplicative homomorphic cryptosystem we present a way to utilize the e-voting in cloud environment. In our future work we would love to have biometric method to authenticate the voter at the top most phase in our algorithm.

REFERENCES

- [1] Yingming ZHAO, Yue PAN, Sanchao WANG et al., "An anonymous voting system based on homomorphic encryption", Communications (COMM) 2014 10th International Conference on. IEEE, pp. 1-4, 2014.
- [2] Francesc SEBE, Josep M. MIRET, Jordi PUJOLAS et al., "Simple and efficient hash-based verifiable mixing for remote electronic voting", Computer communications, vol. 33, no. 6, pp. 667-675, 2010.
- [3] Htet Ne oo, Aye Moe Aung et al., "A Survey of Different Electronic Voting Systems", International Journal of Scientific Engineering and Technology Research Volume.03, IssueNo.16, July-2014, Pages: 3460-3464